

Aditya Shibu

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Dubai, UAE

SUMMARY

An experienced Robotics and Autonomous Systems Developer with a proven track record with hands on projects in fields of AI, ML, Computer Vision, and Control Systems. Enjoys solving real-world problems by combining modern software tools with emerging technologies in automation, robotics, and machine learning.

EDUCATION

- Bachelor's of Science (Hons.) - Computer Science with specialization in AI** **GPA: 4.0/4.0**
Heriot Watt University, Dubai, UAE *Sept 2023 - March 2026*

RESEARCH AND PUBLICATIONS

- Simulating an Autonomous System in CARLA Using ROS2** **Research Paper** - In Proceedings
International Conference on Signal Processing and Information Security (ICSPIS) *November 2025*

WORK EXPERIENCE

- AimbeyondD, Heriot Watt University** **Dubai, UAE**
Research Intern *Nov 2025 - Present*
 - Developing **SkyScript**, a novel real-time framework for natural language control of drone swarms using Large Language Models
- Atlas Racing FS** **Dubai, UAE**
Lead Autonomous Systems Developer *May 2025 - Present*
 - Developed end-to-end autonomous vehicle simulation pipeline in **CARLA**, implementing **deep neural networks for perception** and collecting **11.5K+ training images** from FOSCO dataset for real-time inference optimization
 - Engineered path planning algorithms (**Pure Pursuit, MPC**) integrated with **ROS 2 Humble**, achieving seamless simulation-to-reality transfer for Formula Student AI competition
 - Built comprehensive perception pipeline using **ZED2i stereo cameras and RoboSense Helios 16 LiDAR**, implementing **DBSCAN clustering** and sensor fusion for object detection and localization
 - Optimized **YOLO inference with TensorRT acceleration**, reducing latency by **88%** (140ms → 16ms) and achieving stable **40-50 FPS** for real-time autonomous vehicle operation
 - Led technical development resulting in **1st Place Overall Statics** and **1st Place Real World AI**, demonstrating production-ready autonomous systems performance
- EVCharge** **Dubai, UAE**
Control Systems Developer *Oct 2025 - Oct 2025*
 - Developing control systems to **remote control a Tesla Model Y 2022** remotely over **5G Cellular** using **Comma AI** as the control interface
 - Working on modifying an existing branch of **SunnyPilot** to implement receiving control commands using a controller
 - Achieved remote operation using a wired controller, shifting the focus to remote control over cellular as the next step
- Creed&Bear** **Dubai, UAE**
Quantum Computing Intern *January 2025 - June 2025*
 - Developed **ML models** for Real-life prediction and analysis
 - Optimized the models and algorithms to run on **Quantum Circuits**
- Clarus24 IT Solutions** **Dubai, UAE**
Frontend Developer *August 2024 - October 2024*
 - Developed and maintained fully responsive **ReactJS** website using **Vite**, optimizing build performance and reducing initial load times by **40%** through code splitting and lazy loading
 - Created custom Lottie animations across website pages, improving user engagement while maintaining **98% mobile responsiveness** scores
 - Built **26 reusable React components** including complex mega menus and stateful headers/footers, reducing development time by **30%** and ensuring consistent UI across all pages
- Fincasa Group of Companies** **Dubai, UAE**
AI and Automation Intern *May 2024 - July 2024*
 - Developed and maintained IT infrastructure for **50+ employees**, achieving **99.5% system uptime** and reducing IT support tickets by **40%**
 - Implemented a custom workflow system using **Microsoft Platforms (Teams, Planner, Power Automate, PowerBI)**, streamlining **over 15 business processes** and reducing task completion time by **35%**
 - Built a **WhatsApp Automation and sorting System** using **Python**, reducing manual file sorting time by **75%** and processing **500+ files daily** with around **85% accuracy**
 - Developed a fully functional E-Commerce website using **Wix Studio**, delivering with **sub 3 second load times** and mobile-responsive design

PROJECTS

- **Hydrakon (Formula Student Driverless)**: Worked with a team of 5 to build an autonomous racing car to race at the Formula Student UK | **Tech: YOLO, ROS2, Cartographer, Python, C** (April 2025) | <https://github.com/FS-AI-HWUD>
- **PowerHouse**: Developed a fully-fledged all-in-one smart home application with simulations of smart home devices to control devices that connect over Wi-Fi | **Tech: ReactJS, Firebase, and Python with FastAPI** (April 2025) | <https://github.com/PowerHouse-Project>
- **Local Retrieval Augmented Generation (RAG)**: Developed a fully local RAG with Ollama and LangChain | **Tech: Python, Ollama, and LangChain** (January 2025) | <https://github.com/jonathanjthomas/GDG-RAG-Demo>
- **Space Debris Classification**: Developed a space debris classification model using YOLO, enhancing training data by combining datasets from Roboflow for broader coverage | **Tech: Python, Jupyter Notebook, YOLO** (December 2024) | <https://github.com/adityashibu/spaceDebrisDetection>
- **Personal Portfolio**: Developed my own personal portfolio website using React and NextJS to showcase my skills in a more visually appealing format | **Tech: NextJS, ReactJS, GitHub API** (September 2024) | <https://adityashibu.vercel.app>
- **Heriot Hive**: Created a real-time chat application with User Auth, Express middleware, Database connections, and password encryption | **Tech: MongoDB, ExpressJS, ReactJS, NodeJS, WebSockets** (September 2024) | <https://github.com/adityashibu/HeriotHive>

AWARDS AND CERTIFICATIONS

- **Babcock FS-AI Overall Statics Award Winner** - IMechE - July, 2025
- **FS-AI Real World Award Winner** - IMechE - July, 2025
- **Deep Learning Essentials Intermediate** - IBM - January, 2025
- **GitHub Foundations certification** - GitHub - December, 2024
- **Foundational C# with Microsoft** - FreeCodeCamp - June, 2024
- **Software Engineering Job Simulation** - J.P. Morgan - June, 2024
- **Google Analytics Individual Qualification** - Google - July, 2023
- **CS50P - Introduction to programming using Python** - EdX - July, 2023
- **Trinity Music Certification** - Trinity College London - March, 2014

LEADERSHIP AND VOLUNTEERING EXPERIENCE

- **Heriot Watt University** Dubai, UAE
MACS Department Officer *August 2025 - Present*
 - Lead and coordinate a team of **12 class representatives** across all Mathematical and Computer Sciences courses (Years 1-4), facilitating effective communication and problem resolution, serving as primary student advocate in faculty committees and departmental meetings, successfully addressing academic concerns and implementing policy changes that impact **600+ MACS students**
- **Google Developer Groups** Dubai, UAE
Lead of AI and ML *May 2025 - Present*
 - Promoted from Co-lead to Lead after organizing a successful workshop with **120+ attendees**, demonstrating exceptional event management and technical presentation skills
- **WattBuds - Peer Mentoring program** Dubai, UAE
Event Coordinator *September 2024 - March 2025*
 - Co-organized record-breaking campus event with **215+ attendees**, surpassing all previous university events and setting new benchmark for student participation and engagement
- **COP28** Dubai, UAE
IT Volunteer *December 2023*